

[INTERNAL WORKING DOCUMENT
/NOT FOR DISTRIBUTION]

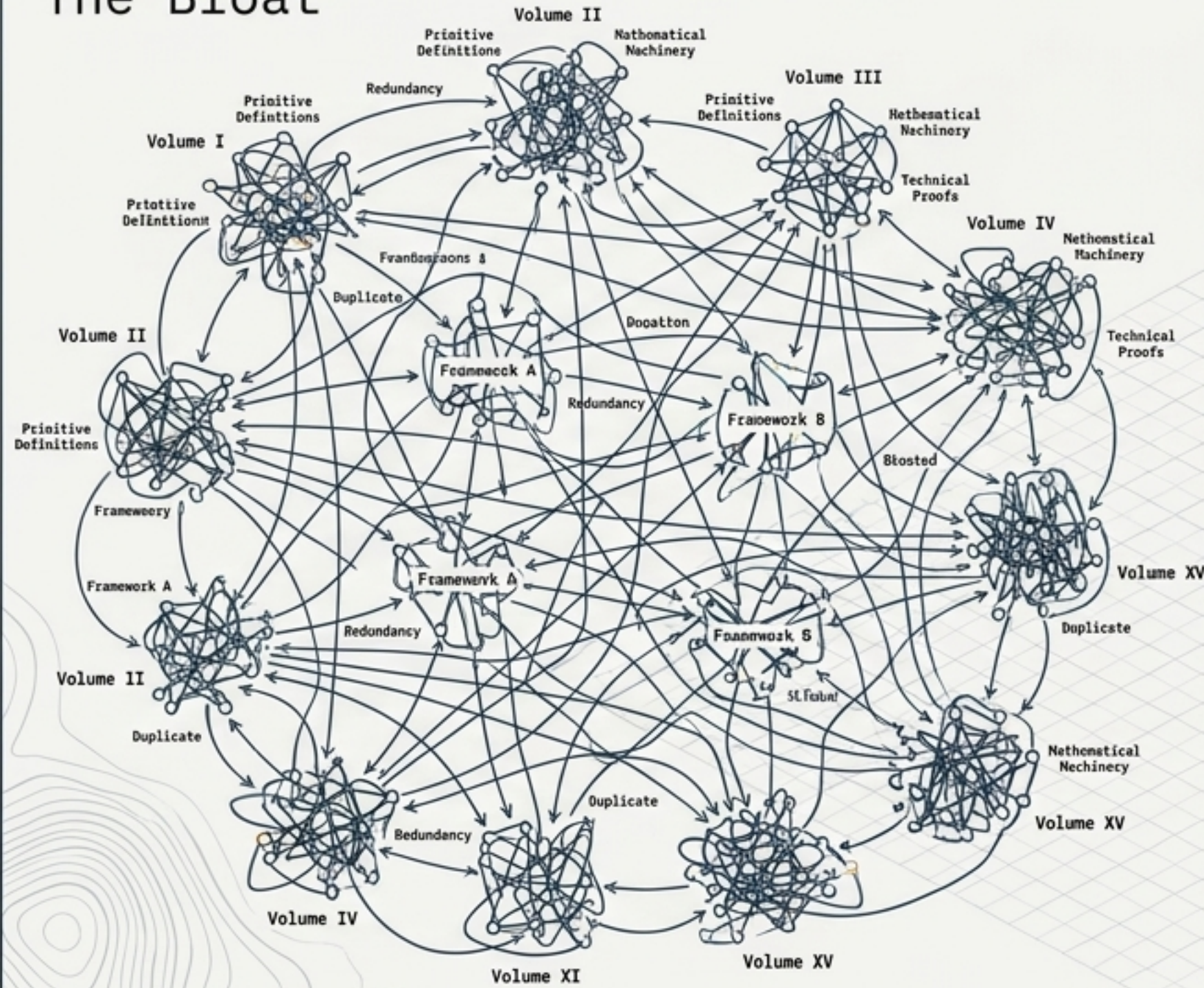
The Flyxion Dependency Atlas

One Variational Principle. Many Instantiations.

An architectural blueprint for collapsing the 15-volume corpus into a unified mathematical structure.

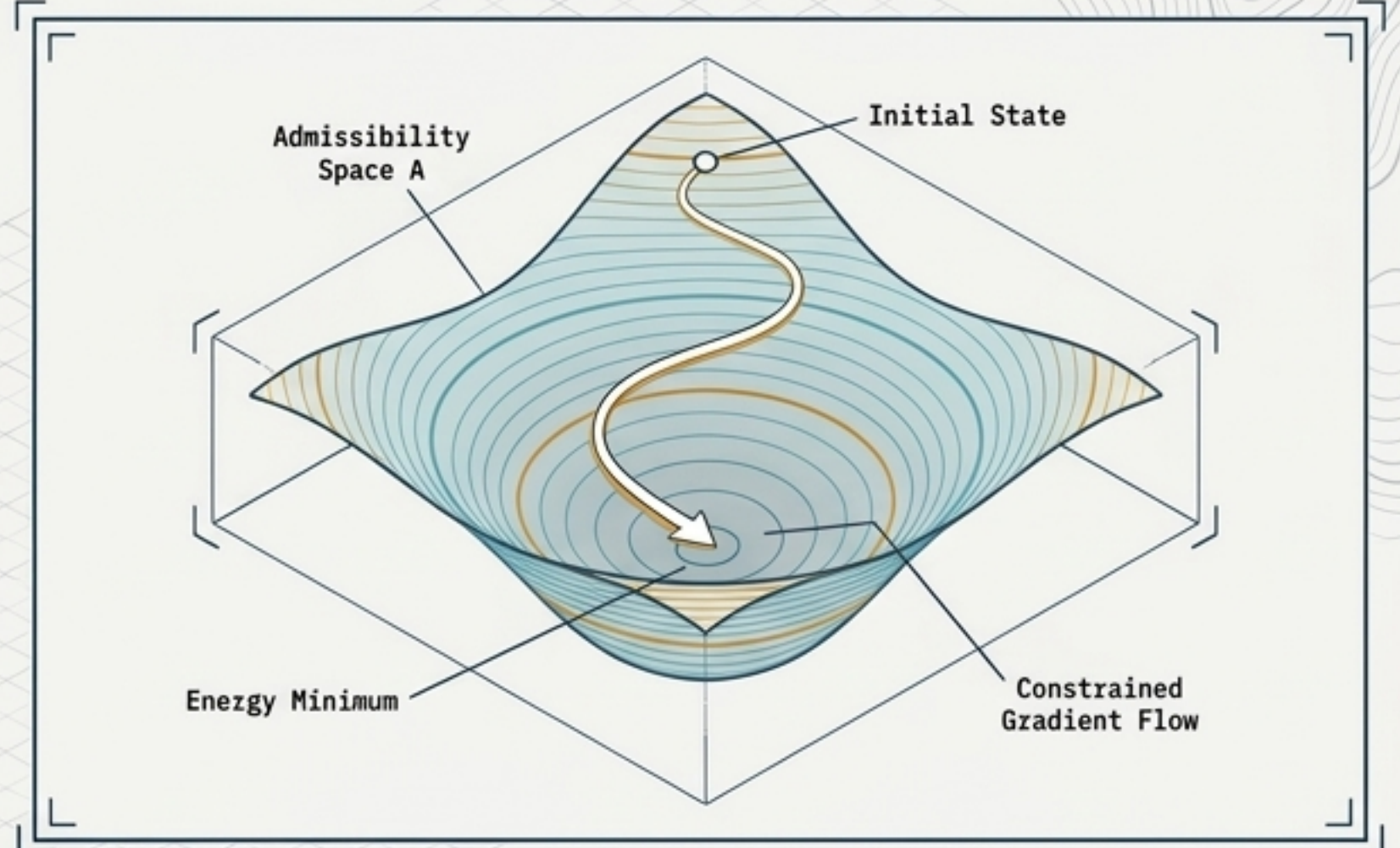
The Master Claim: Fifteen Volumes, One Principle

The Bloat



The current curriculum proposes 15 distinct frameworks. This creates massive redundancy in primitive definitions, mathematical machinery, and technical proofs.

The Reality

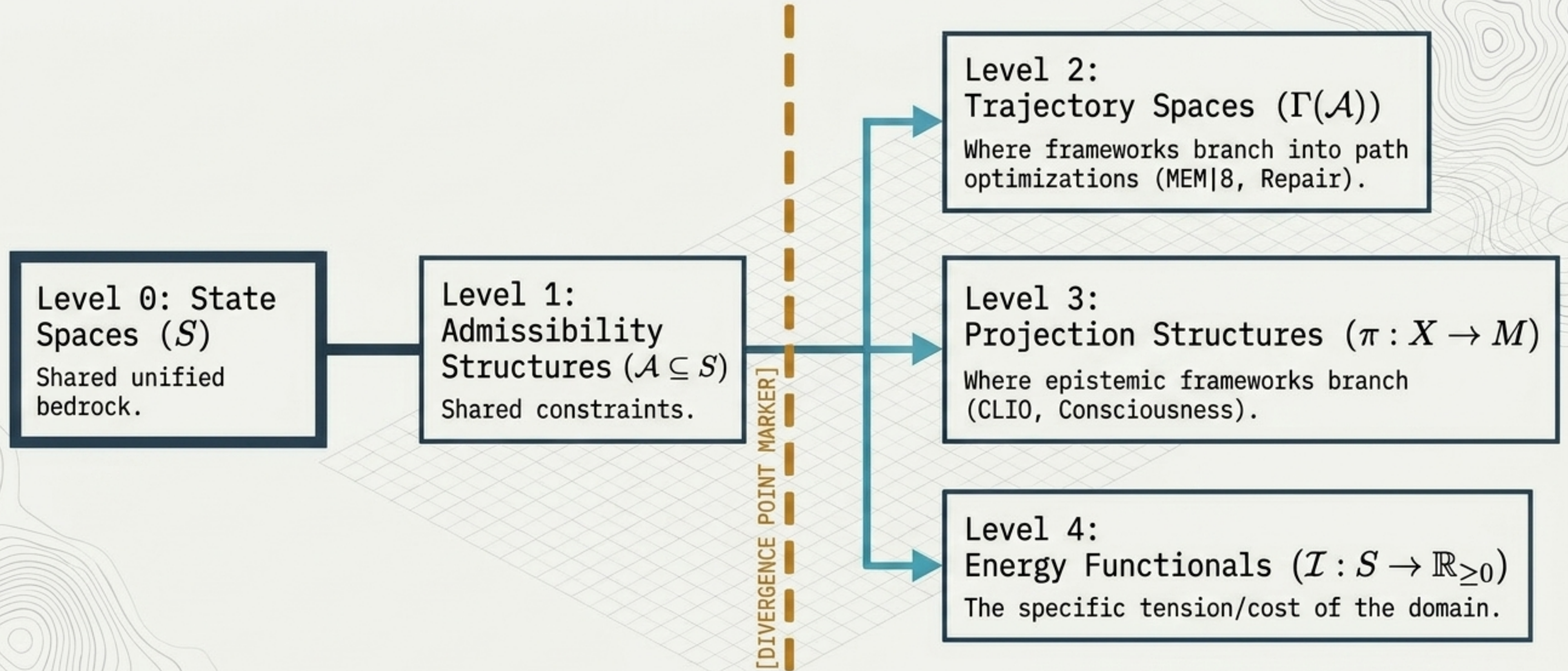


All Flyxion frameworks are mathematically identical at their core. They are all constrained gradient flows on admissibility spaces.

$$\frac{dx}{dt} = -\nabla_g \mathcal{I}(\mathbf{x}) + \xi_t$$

Different frameworks are simply different energy landscapes evaluated under different constraints.

The 5-Level Structural Hierarchy



Takeaway: Frameworks differ strictly at Levels 2-4. They share Levels 0-1 entirely.

The Missing Primitive: Admissible Trajectory γ

Definition: A continuous path $\gamma : [0, T] \rightarrow \mathcal{A}$ where velocity $\dot{\gamma}(t)$ is tangent to the boundary of $\mathcal{A}$.

MEM|8: Memory Trace $\mu = (e_1, \dots, e_n)$
— Discrete trajectory in event space.

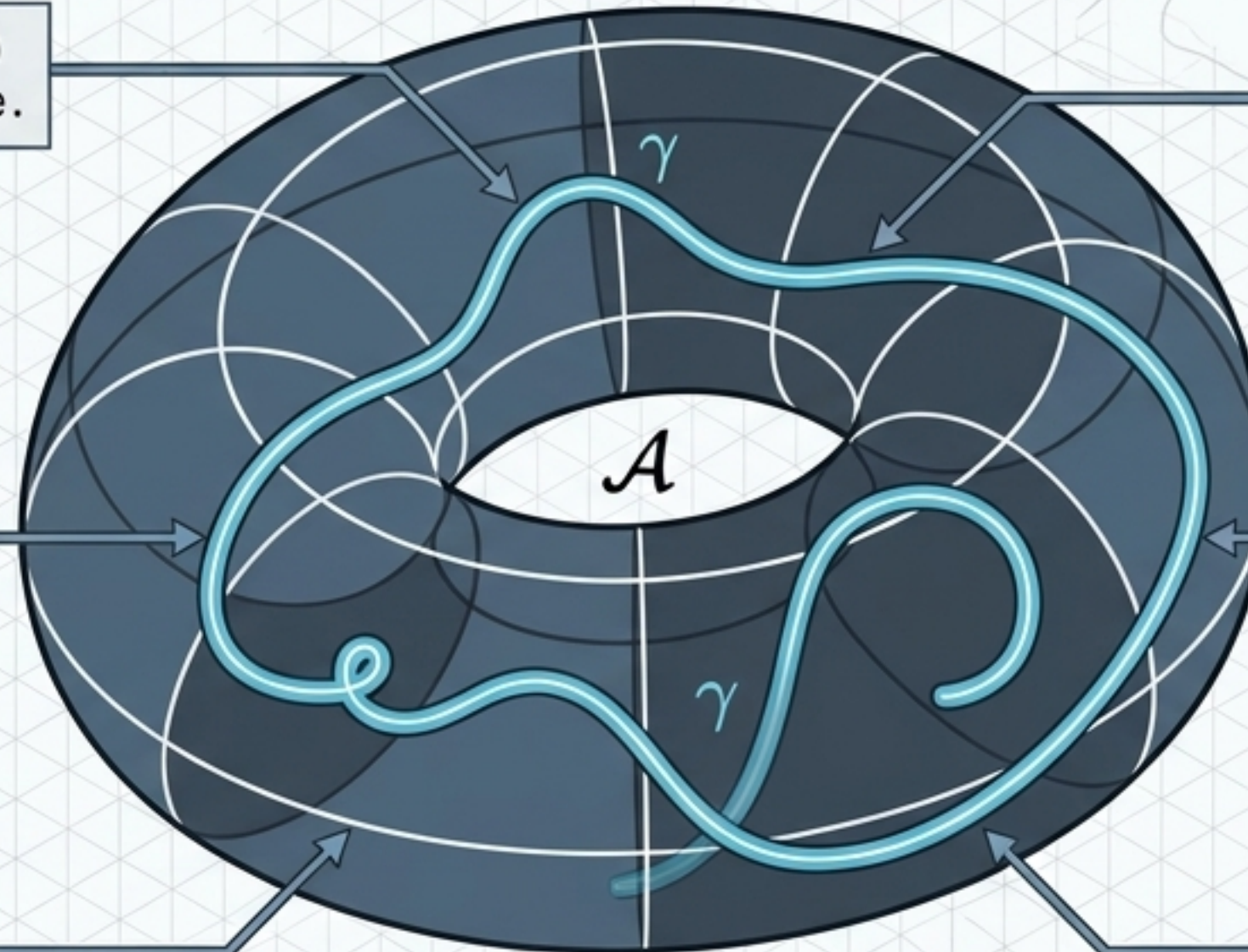
Repair Theory: Repair Path
— Trajectory in $\mathcal{A}$ reducing divergence.

Simulated Agency: Future Path
— Trajectory maximizing expected reachability $\mathbb{E}[\Omega(\gamma(T))]$.

Ecphory: Cognitive Path
— Wave from present state to target memory.

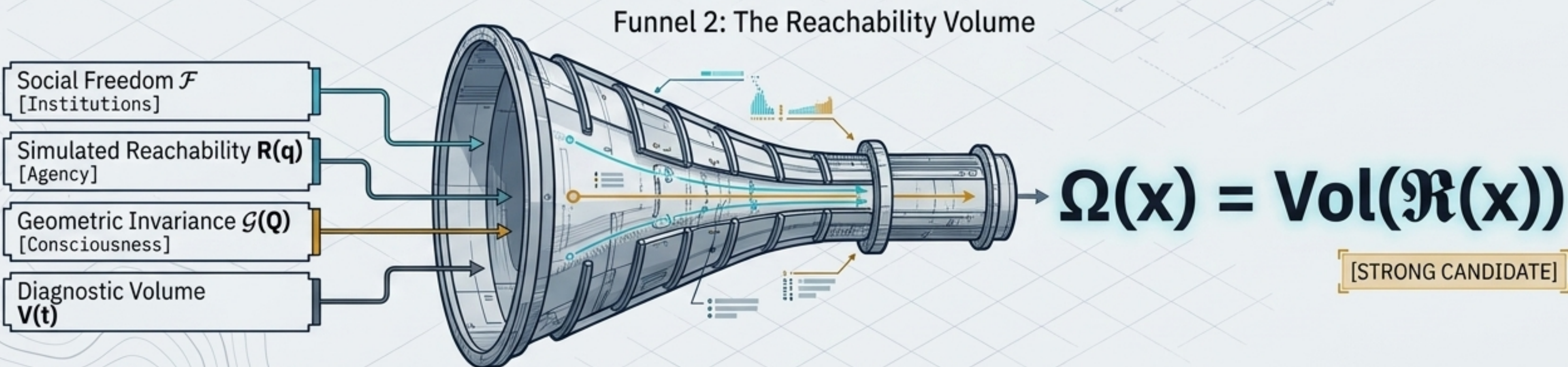
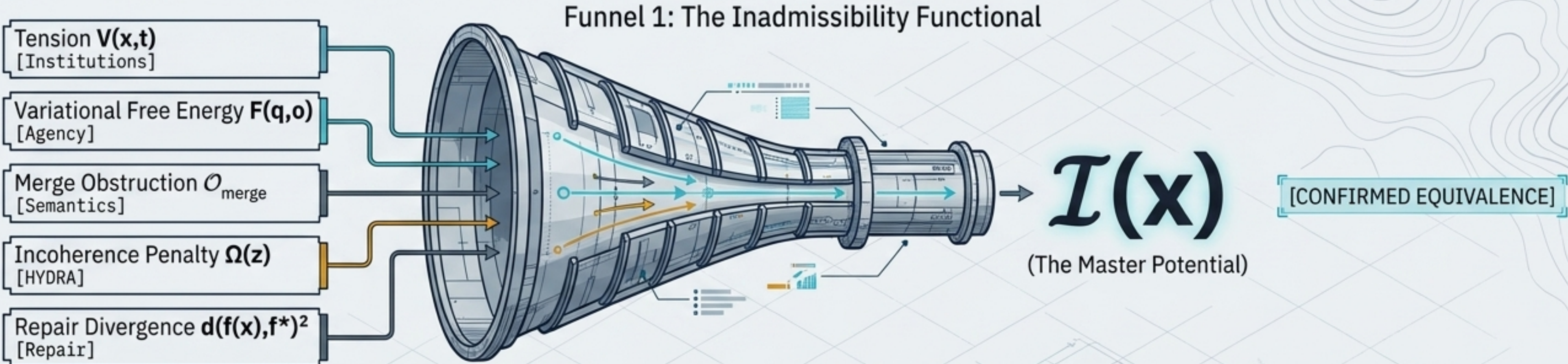
Yarncrawler: Semantic Route
— Trajectory completing a partial world-state.

TARTAN: Preserved Path
— Trajectory preserved under decomposition.

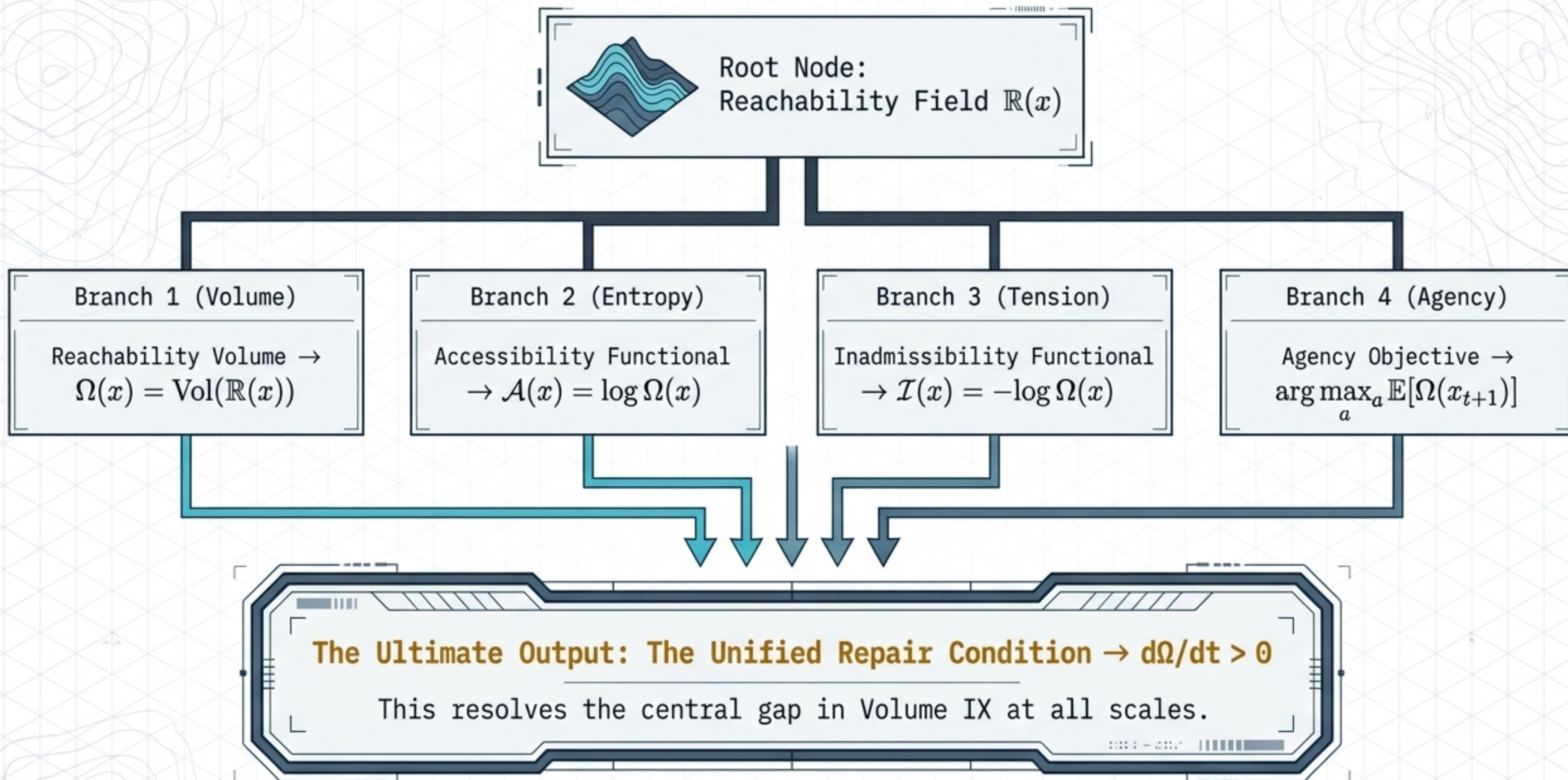


Takeaway: The missing primitive, the **Admissible Trajectory** γ , is the unifying concept connecting various higher-level frameworks.

The Equivalence Collapse: Eradicating Redundant Primitives

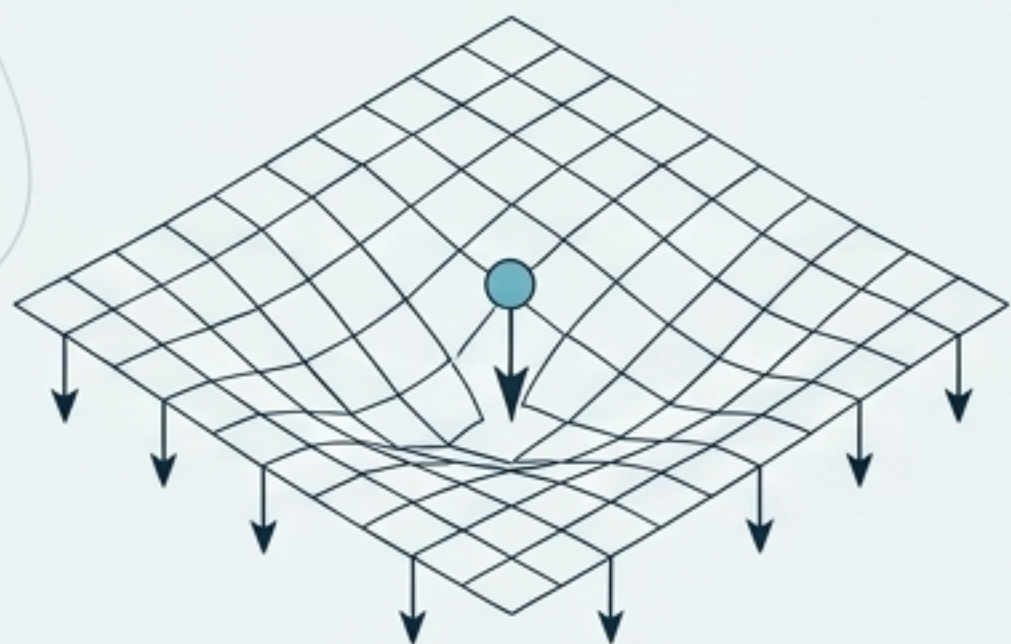


The Reachability Field $\mathbb{R}$: A Proposed Meta-Primitive



Paradigm Shift: State Functionals vs. Path Actions

STATE FUNCTIONALS

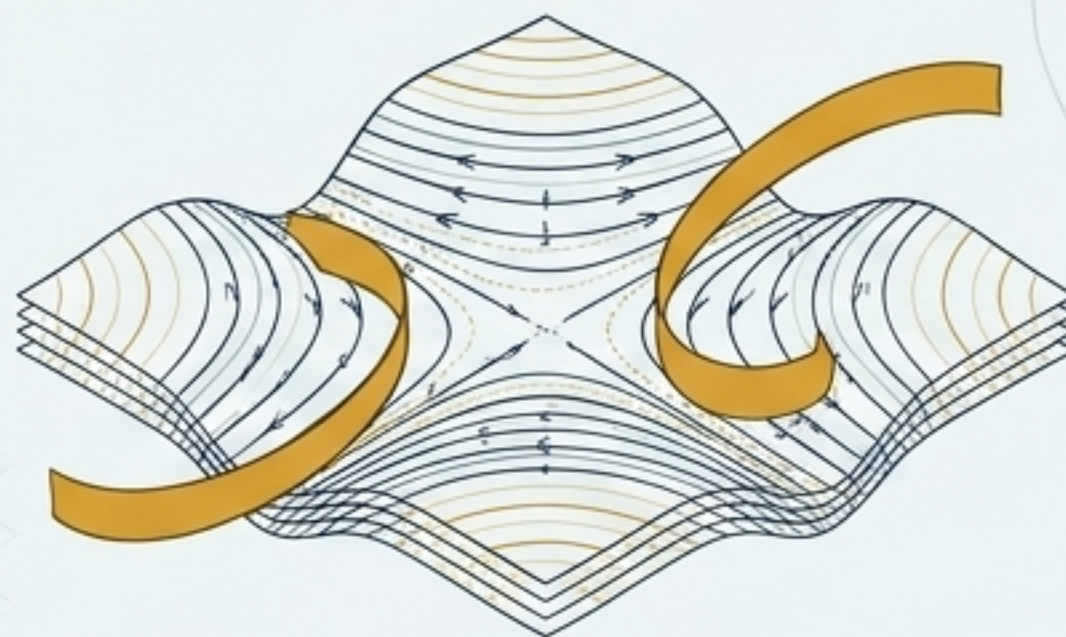


$$\dot{x} = -\nabla_g \mathcal{I}(x)$$

Concept: Assigns cost to a *point*.
Instantaneous cost reduction.
No memory.

Frameworks: RSVP, CLIO.

PATH ACTIONS



$$\mathcal{J}[\gamma] = \int_0^T \mathcal{L}(\gamma(t), \dot{\gamma}(t), t) dt$$

Concept: Assigns cost to an *entire trajectory*.
Global path optimization. Encodes
history and irreversibility.

Frameworks: Spherepop, MEM|8, Agency, Repair,
Ecphory, Yarncrawler, TARTAN.

The Revised Master Claim: The Path Lagrangian

$$\gamma^* = \arg \max_{\gamma \in \Gamma(\mathcal{A})} \mathcal{J}[\gamma]$$

Annotation 1:

The globally optimal admissible trajectory.

Annotation 2:

The trajectory space bounded by constraint geometry.

Annotation 3:

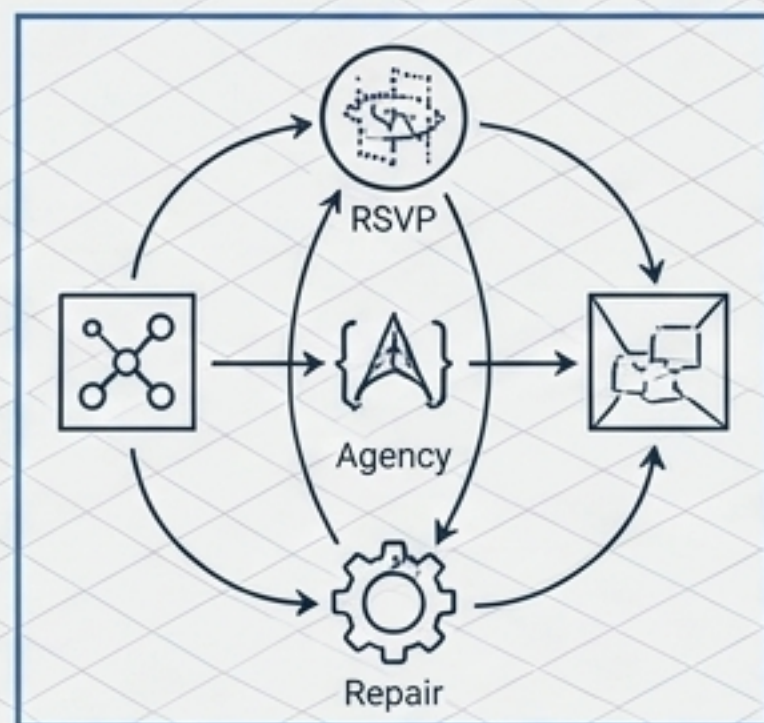
The Path Lagrangian. More general than a state potential.

Callout Block

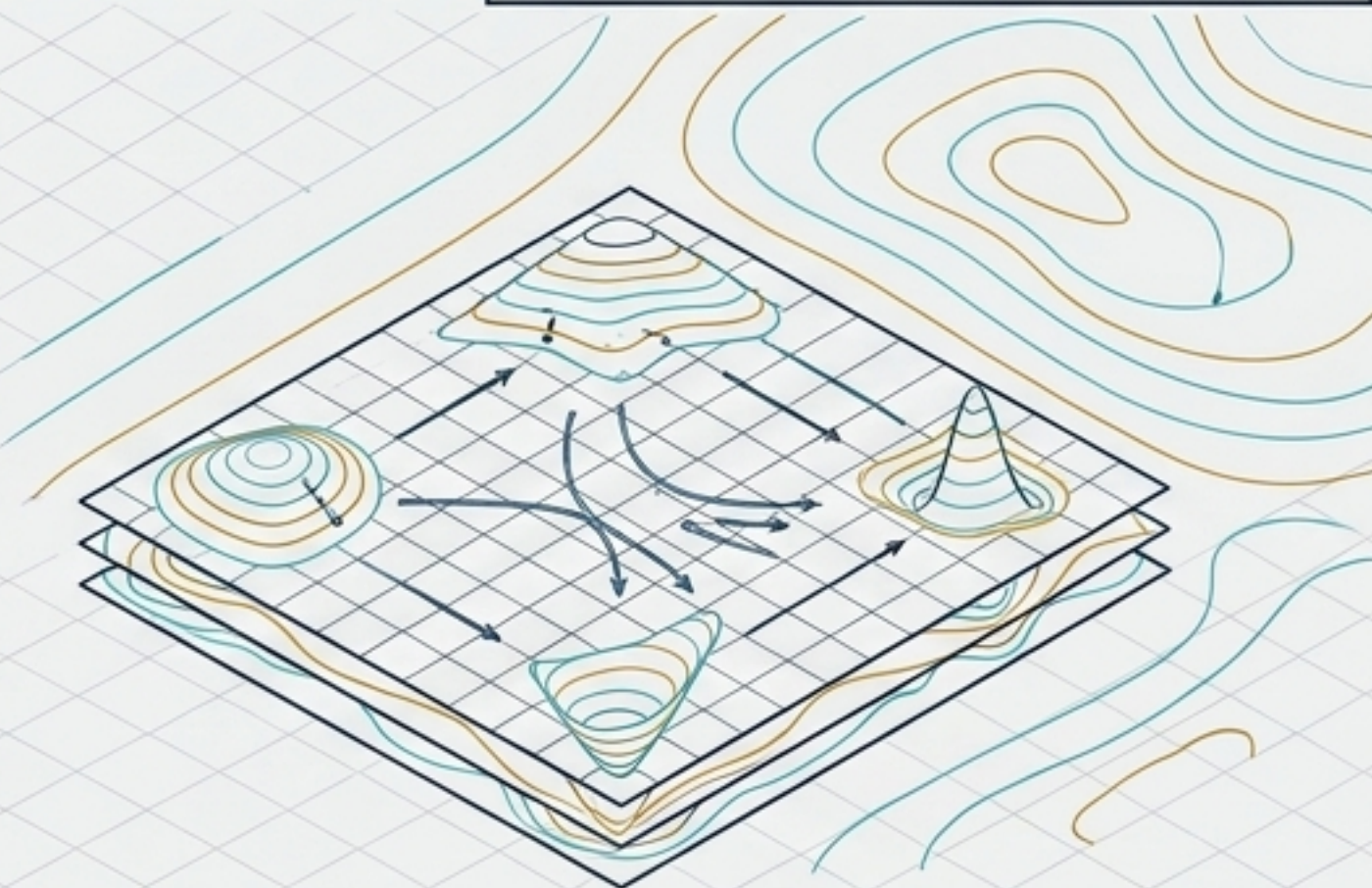
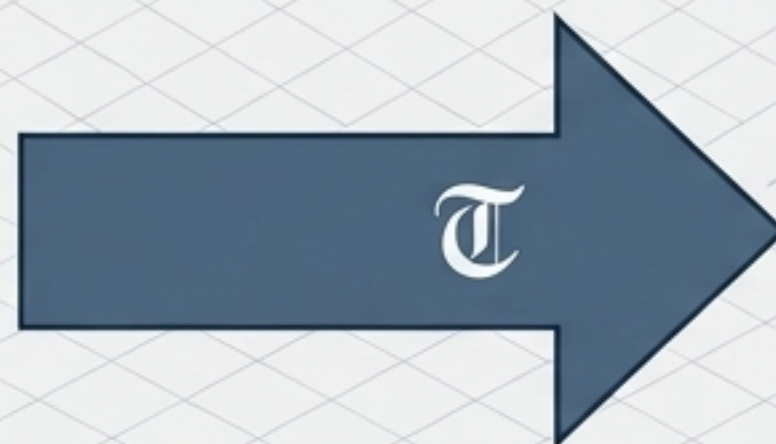
The original gradient flow ($\dot{x} = -\nabla I$) is merely a special case where $\mathcal{L} = \frac{1}{2}g(\dot{\gamma}, \dot{\gamma}) + \mathcal{I}(\gamma)$. The variational form accommodates both kinetic potentials and path-level irreversibility.

The Ultimate Unifier: The Energy Functor $\mathcal{T}$

[THE VOLUME XV
SYNTHESIS THEOREM]



Frameworks



Energy Landscapes

1 The Theorem

There exists a functor
 $\mathcal{T} : \mathbf{Frameworks} \rightarrow \mathbf{EnergyLandscapes}$

2 The Mapping

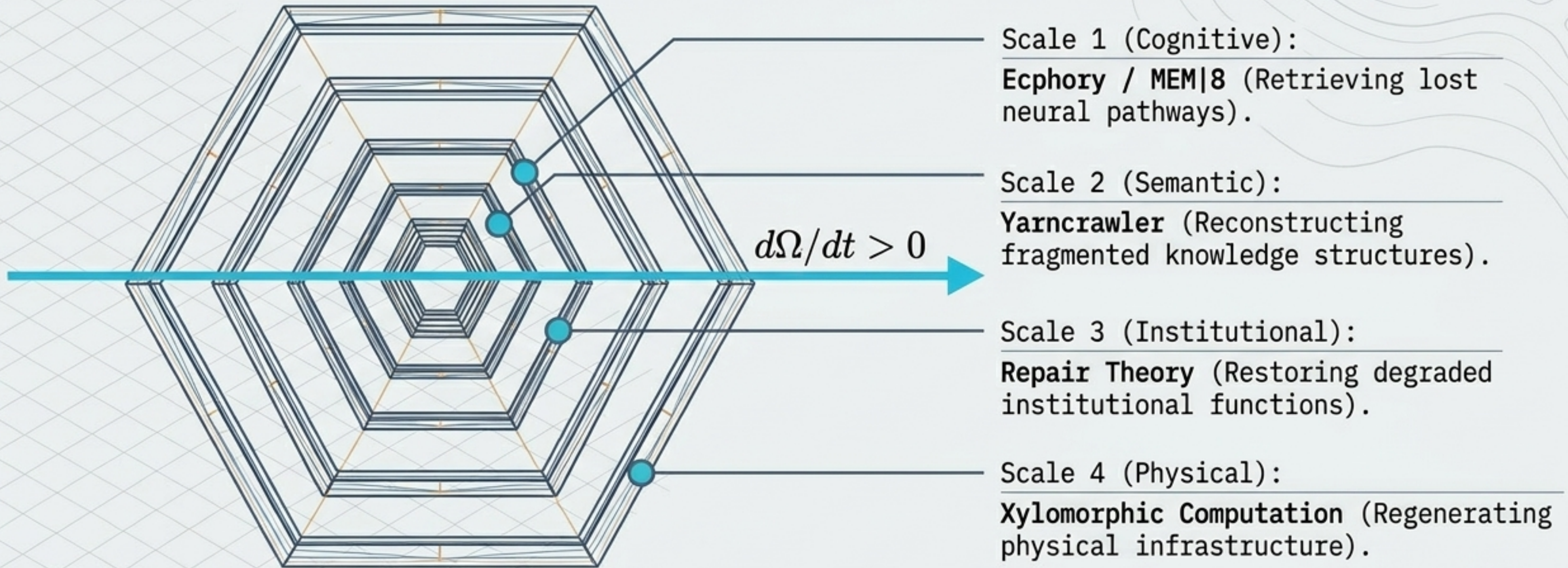
$$\begin{aligned}\mathcal{T}(\text{RSVP}) &= V \\ \mathcal{T}(\text{Agency}) &= F \\ \mathcal{T}(\text{Repair}) &= d^2\end{aligned}$$

3 The Implication

"Are these the same framework?" is no longer a philosophical debate. It is a rigorous mathematical question of whether
 $\mathcal{T}(\mathcal{F}_1) \cong \mathcal{T}(\mathcal{F}_2)$.

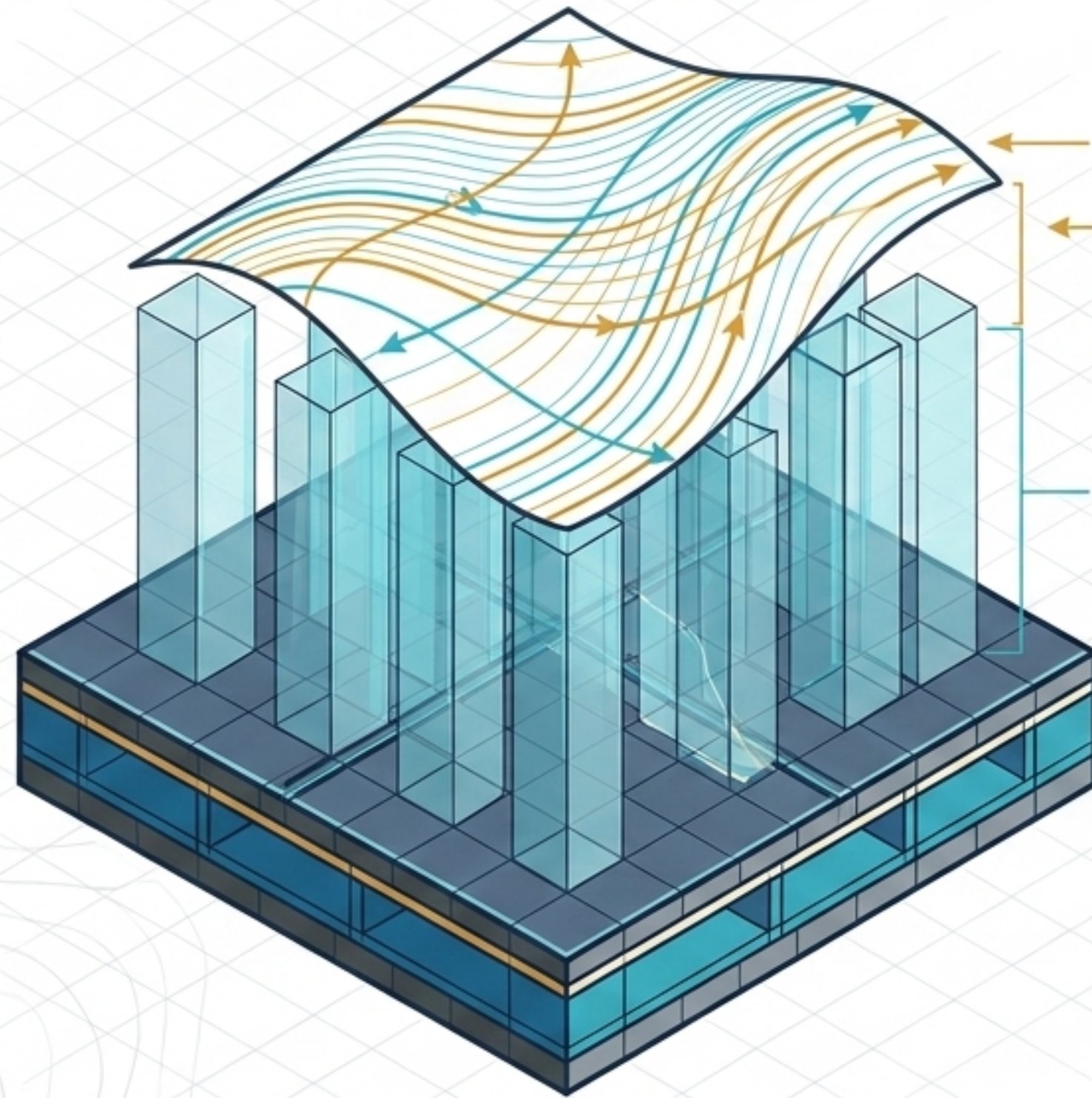
Structural Fallout: The Elevation of Repair Theory

Repair is not a downstream application. It is a transformation class occurring at every scale of the corpus, governed by a single morphism $\mathcal{R}_\epsilon$.



Takeaway: Volume IX (Repair Theory) must be elevated from the Application tier to the Core Tier.

The New Paradigm: The Three-Layer Architecture



Layer 3

Layer 3: Trajectory Theory
(How do we move, remember, and repair?)

Core Objects: Trajectory space $\Gamma(A)$,
Path Lagrangian $J[\gamma]$. (Vols IV - XII)

Layer 2

Layer 2: Projection Theory
(What do we observe?)

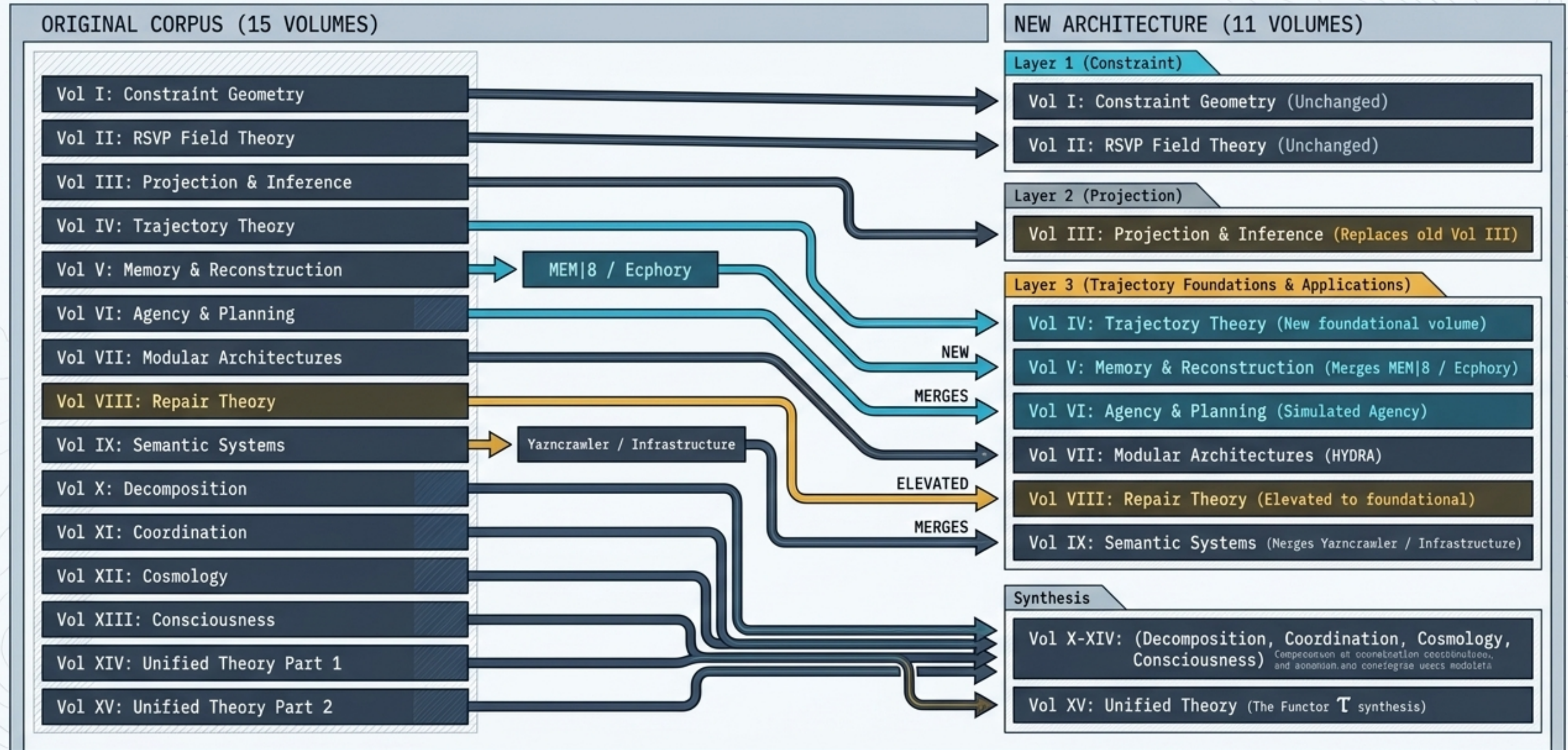
Core Objects: Projection operator $\pi: X \rightarrow M$,
compression, observation. (Vol III)

Layer 1

Layer 1: Constraint Geometry
(Where can we be?)

Core Objects: Admissibility spaces A ,
metrics g , inadmissibility I . (Vols I, II)

Restructuring the Corpus: From 15 Volumes to 11



The Immediate Prerequisite: Volume 0

Before drafting begins on the new 11-volume architecture, an internal consistency check (40-60 pages) must be executed.

dossier checklist



Canonical Symbols Table: Standardize $\mathcal{A}$, π , $\mathfrak{R}$, and $\mathcal{I}$ across all domains. Stop using different letters for the same math.



Proof Status Register: Formal statement of the 11 asserted-but-unproved foundational gaps (e.g., Reachability Principle, Spherepop Confluence).



Master Claim Formalization: Complete the proof sketch for $\gamma^* = \arg \text{ext } \mathcal{J}[\gamma]$.



Equivalence Dossiers: One-page formal geometric arguments confirming the 7 structural equivalences.

When Volume 0 is secure, we do not have a library of theories. We have a unified academic field.